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Breaking and Keyword Matched News

Partnering through PACE with



MACQUARIE
University

The Client



Background

— — —



Founded in **2005** by **Dr. Hong Liang Qiao**
Sydney, Australia

What is Lexxe?

— — —



An internet **Search Engine**.

But **NOT** just another **G**gle**...

What makes it different?

— — —



Sentiment based **Search** Technology.

A “**next generation**” search engine.

Imagine ...



— — —

... having the capability to assess **public opinion** of **anything** across the **globe** with just a few keystrokes.

Anything...?



A **brand**, a **product**, a **person**, a **service**, an
organization, an **event**, a **concept**...

ANYTHING or **ANYONE**

(yes... even notoriously popular U.S. Presidents)

The outcome...

— — —



Search results delivered with **sentiment score**
aggregated from **millions** of **websites**.

This is significant...

— — —



Sentiment helps with **judgement** and ultimately
better decision making.

The Project



Background

— — —



Lexxe offers a mobile app named “**News and Moods**” enabling **news search** and **sentiment shift** identification.

Background

— — —



Provide **customers** with a **news** article **feed**.

Objective

— — —



Identify unique **breaking** and **keyword** matched
news articles.

Scope



A **systematic process** that **iterates** a given set of **news articles** and **returns** a **reference** for **articles** meeting **predefined criteria**.

Keyword Matched News



Keyword selection

Goal:

- Find news articles that contain specified keywords
- It works for any number of keywords

Vladimir **Putin**: Lifting Russia **sanctions** would benefit all

Vienna: Western **sanctions** against Russia haven't worked and both Moscow and the West would benefit from lifting them, Russian President Vladimir **Putin** declared on Tuesday as he visited Austria - one of the European Union's more Moscow-friendly members.

Putin said the restrictions are "harmful for everyone - those who initiated them and those who are targeted by them."

The United States, the European Union and other Western allies imposed a slew of **sanctions** against Moscow over its 2014 annexation of Ukraine's Crimean Peninsula and its support for separatists in eastern Ukraine who are fighting the government.

Example SMH article containing "putin" and "sanctions"

Requirements from Lexxe



- Return **at most 6 articles** per **day**
- Do not push **duplicate** articles
- Format: *article.docid;'Keyword';reason;pub_keywords;title*

Example

4651424000744407825;Keyword;0.30-Woolworths;Woolworths;
Brisbane woman claims she's found a bullet in her Kangaroo
steak

Solution



Keyword news

- Search for keywords in title and body of article
 - Identify the number of keyword matches
 - Add matches to a total score for the article
- If score > threshold, push that article

Demo



Keyword Matched News Demo

Breaking News



Requirements Breaking



What is Breaking News:

- Initial reports with basic description of news
- Usually brief articles
- More details of incident are expected to follow OR
- The reported incident is still developing

Requirements from Lexxe



- Articles sent truly represent breaking news
- Push notification within 1 hour of first detected article
- No more than 6 items pushed per day
- No duplications of the same story

Using Sindex to detect sudden increases



Title: Carbon emissions: Artificial intelligence used to fight deforestation in Congo

Sindices: { **Elizabeth Goldman** : 1, **Paris Agreement** : 0, **Luxembourg** : 0, **Congo** : 2, **Amazon** : 0, **WRI** : -1 }

Finding Related Articles



Other Sindices: {**Paris Agreement** : 0, **Congo** : 2, **Amazon** : 0}

Jaccard similarity: **0.5**

- Simple and works well for similar sized sets

Cosine similarity: **0.717**

- Fast and works well regardless set size.

Breaking News

Demo

Results



Keywords

● CBA	8 out of 15 correct	- 53 % Accuracy
● China	12 out of 18 correct	- 66 % Accuracy
● Google	8 out of 15 correct	- 53 % Accuracy
● Malcolm Turnbull	4 out of 5 correct	- 80 % Accuracy
● Woolworths*	2 out of 8 correct	- 25 % Accuracy

- * Only 2 out of 4 days with important news
- * System correctly ignored all non-important articles
- * Only 4 important news stories in total

Breaking News



Worst Case 18 out of 27 correct – 67% Accuracy

Best Case 24 out of 27 correct – 88% Accuracy

- 6 news articles output by system that may still yet be re-classified as breaking news
- Similar or same article not returned more than once

Reflection



Reflecting on the project (learning outcomes)?



- Working as a group
- Version control for software dev
- Documentation of a project
- Working with a client
- Working on a “real life” problem

What's Next?



Future Improvements



- Refine the model even further
- More test data
 - Using social media to generate test data
 - Broader global dataset
- Advanced keyword filtering
 - E.g. Trump, Donald Trump, POTUS are the same people
- Improve efficiency for big data
 - processing a global datastream of news articles

Thank You

